

A REPORT ON
SMARTTRIP AI: AN INTELLIGENT TOURISM PLANNING
AND RECOMMENDATION SYSTEM

Submitted by
E Vidya - 20241CSE1084

Under the guidance of,
Mr.Aadil Ferooz
Ms.Chinmayi G V
in partial fulfillment for the award
of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
AT



PRESIDENCY UNIVERSITY
BENGALURU
AUGUST 2026

PRESIDENCY UNIVERSITY
PRESIDENCY SCHOOL OF COMPUTER
SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the report of **CSE7000 -- Internship** on “**SmartTrip AI: An Intelligent Tourism Planning and Recommendation System**” being submitted by **E Vidya** bearing roll number **20241CSE1084** in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** is a bonafide work carried out under our supervision.

Mr.Aadil Ferooz
Designation
PSAI-AC
Presidency University

Ms.Chinmayi G V
Designation
PSAI-AC
Presidency University

Ms. Vidhya Rengasamy
Internship coordinator PSAI-AC
Presidency University

Dr. Bhuvaneshwari Patil
School level Internship
coordinator PSAI-AC
Presidency University

Dr. Nagaraja S R
Head of the Department
PSAI-AC
Presidency University

Dr. Duraipandian N
Dean -- PSCS & PSIS
Presidency University

PRESIDENCY UNIVERSITY

**PRESIDENCY SCHOOL OF COMPUTER
SCIENCE AND ENGINEERING**

DECLARATION

I hereby declare that the work, which is being presented in the report entitled “**SmartTrip AI: An Intelligent Tourism Planning and Recommendation System**” in partial fulfilment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering**, is a record of my own investigations carried under the guidance of **Ms.Chinmayi G V** and **Mr.Aadil Ferooz** , Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

E Vidya-- 20241CSE1084

ACKNOWLEDGEMENTS

First of all, I am indebted to the GOD ALMIGHTY for giving me an opportunity to excel in my efforts to complete this internship on time.

I express sincere thanks to our respected Dean **Dr. Duraipandian N**, Presidency School of Computer Science and Engineering & Presidency School of Information Science, Presidency University for getting me permission to undergo the internship.

I express heartfelt gratitude to our beloved **Dr. Nagaraja S R**, Head of the Department, Presidency School of Artificial Intelligence and Advanced Computing, Presidency University, for rendering timely help in completing this internship successfully.

I am greatly indebted to my reviewer **Mr.Aadil Ferooz,Ms.Chinmayi G V** Presidency School of computer science and engineering,for their inspirational guidance, and valuable suggestions and for providing a chance to express technical capabilities in every respect for the completion of the internship work.

I would like to convey gratitude and heartfelt thanks to the Internship Coordinator **Dr. Bhuvaneshwari Patil** and Program Internship Coordinator **Ms. Vidhya Rengasamy**. I thank the faculty and staff members of Presidency University for their support during my Internship. And I also thank my family and friends for the strong support and inspiration they have provided me in bringing out this internship.

E Vidya (20241CSE1084)

ABSTRACT

Tourism planning has become increasingly dependent on digital platforms, where travellers must collect information from multiple sources before deciding on a trip. Choosing a suitable destination, finding accommodation, identifying nearby attractions, estimating travel expenses, and preparing a proper itinerary often require visiting several websites and applications. Since each platform focuses on only one aspect of travel planning, users are forced to switch between different services and manually compare the available options. This process is not only time consuming but also makes it difficult to create a travel plan that matches individual interests, budget, and travel preferences. Most existing travel planning systems rely on keyword searches and fixed filtering options, which often fail to understand the actual meaning of a user's request when it is written in natural language. As a result, users spend additional time refining their searches to obtain useful recommendations.

Smart Trip AI was developed during the AI GPU Summer Internship to address these limitations by providing a single intelligent platform for tourism planning. The objective of the project was to design and implement a recommendation system capable of understanding user preferences and generating personalized travel suggestions without requiring users to search across multiple websites. The application combines Artificial Intelligence, Natural Language Processing, semantic search, and recommendation system techniques to provide destination recommendations, hotel suggestions, nearby attractions, travel budget estimation, and itinerary generation through one integrated interface.

The central component of the application is its semantic recommendation engine. Unlike traditional search systems that compare only exact words or phrases, the proposed system focuses on understanding the meaning and context of user queries. User preferences entered in natural language are converted into dense vector embeddings using the pretrained all MiniLM L6 v2 sentence transformer model. The generated embeddings are compared with embeddings created from the tourism dataset using cosine similarity to identify destinations that best match the user's interests. To improve recommendation quality, the similarity score is combined with additional parameters such as destination ratings, popularity, and budget compatibility before

generating the final ranked results. This approach enables the platform to provide recommendations that are more relevant and personalized than conventional keyword based search methods.

The application consists of five major functional modules that work together to simplify the travel planning process. The Destination Recommendation module analyses user preferences and recommends suitable tourist destinations according to their interests. Once a destination is selected, the Hotel Recommendation module provides accommodation options by considering factors such as destination, hotel ratings, pricing, and user budget. The Nearby Attractions module displays important tourist places, sightseeing locations, and activities available around the selected destination. The Budget Analysis module estimates the expected cost of the trip by calculating expenses related to accommodation, transportation, food, and tourist activities. The final Itinerary module combines information from all the previous modules and prepares a structured day wise travel plan that presents the complete journey in an organized format.

The complete system was implemented using Python, with Streamlit serving as the framework for developing the interactive web application. Pandas and NumPy were used for data cleaning, preprocessing, and analysis of tourism related datasets containing destination details, hotel information, attraction data, ratings, and pricing. The datasets were processed to remove inconsistencies and standardize important fields before being used by the recommendation engine. Each module was designed, implemented, and tested independently before integration, making the application easier to maintain and extend in the future.

The project provided an opportunity to apply concepts learned during the AI GPU Summer Internship in a practical software development environment. Knowledge of Machine Learning, Neural Networks, Convolutional Neural Networks, Transformer models, Large Language Models, the Hugging Face ecosystem, Generative AI, and GPU computing contributed to the development of the application. Instead of building a language model from scratch, the project demonstrates the practical use of pretrained transformer models for semantic recommendation tasks, allowing the system to understand user intent more effectively while reducing computational complexity.

Introduction

Planning a trip can often be a time-consuming and complicated process, as travellers need to search through multiple sources for suitable destinations, tourist attractions, hotels, transportation options, and travel expenses. Since this information is usually scattered across different platforms, it can be difficult for travellers to compare options and create an itinerary according to their interests, available time, and budget. Traditional tourism recommendation systems may also rely heavily on keyword-based searches, which may not accurately understand the actual intent expressed by users in natural language.

To address these challenges “SmartTrip AI – Intelligent Tourism Planning and Recommendation System” was developed during the AI GPU Summer Internship as an AI-powered solution for simplifying the travel-planning process. The system provides a single, user-friendly platform for discovering tourist destinations, generating travel itineraries, recommending hotels, exploring nearby attractions, and estimating travel expenses based on user preferences. The project integrates concepts such as Artificial Intelligence (AI), Natural Language Processing (NLP), data preprocessing, Transformer-based semantic search, and recommendation techniques.

A key component of SmartTrip AI is the pre-trained “all-MiniLM-L6-v2” Transformer model, used through the Sentence Transformers framework to understand the semantic meaning of user queries and tourism information. The system converts text into numerical representations called embeddings and uses cosine similarity to identify destinations that are most relevant to the user's interests. Tourism and hotel datasets are processed and organized to provide information such as destination categories, locations, ratings, hotel details, and prices, enabling the system to generate meaningful recommendations.

The application is developed using Python for data processing and recommendation logic and Streamlit for the interactive web interface. By combining AI-based semantic recommendations with trip planning, hotel discovery, nearby attraction suggestions, and budget estimation, SmartTrip AI aims to reduce the time and effort required for travel planning. Overall, the project demonstrates the practical application of AI and NLP technologies in creating a personalized, intelligent, and user-friendly digital tourism solution.

Objectives

The internship project was carried out with the following objectives:

- Recommend tourist destinations based on user interests and travel preferences
- Generate personalized, destination-specific travel itineraries.
- Suggest hotels and accommodation options suited to user budget and preferences.
- Identify nearby attractions and points of interest for selected destinations.
- Provide travel budget analysis and cost estimation.
- Demonstrate practical, end-to-end implementation of the AI concepts (Machine Learning, Neural Networks, Transformers, and Generative AI) covered during the internship.

Work Progress

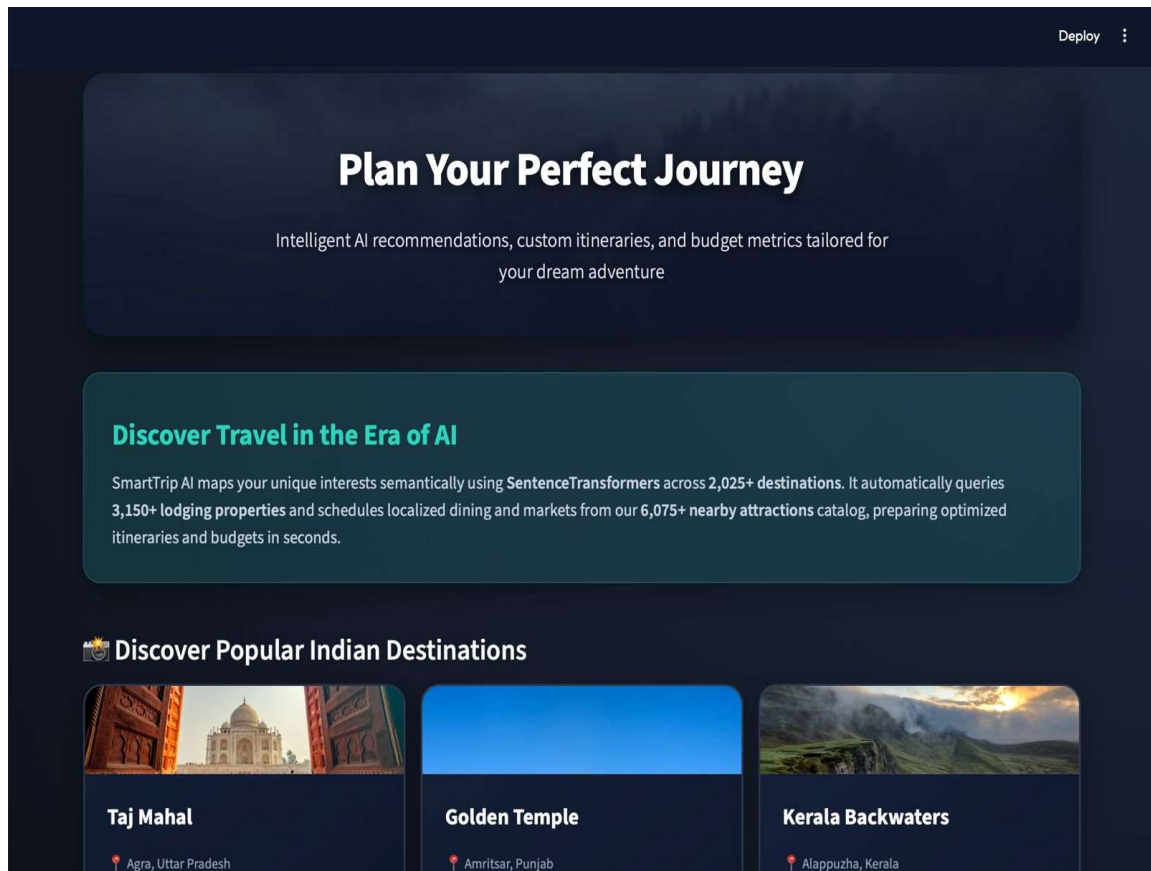
Development proceeded through data collection, system design, and iterative implementation. Tourism, hotel, and attraction datasets were collected and pre-processed to remove inconsistencies and standardise fields such as destination descriptions, ratings, popularity indicators, and pricing, forming the foundation for the recommendation engine.

A modular architecture was designed in which user preferences are converted into semantic embeddings using the transformer-based all-MiniLM-L6-v2 sentence embedding model. The recommendation engine compares these embeddings against the processed tourism dataset using cosine similarity, and the resulting destinations are ranked using a combination of semantic similarity, ratings, popularity, and budget compatibility. Ranked destinations are then passed through dedicated hotel-recommendation, nearby-attraction, and budget-analysis modules before being presented on a unified dashboard.

The application was implemented in Python, using Streamlit for the interactive front end and Pandas/NumPy for data processing. Each module -- recommendation engine, semantic search, hotel recommendation, nearby attractions, and budget analysis -- was built and tested independently before integration into the final dashboard flow (Home Page → User Input → Recommendation Engine → Destination Selection → Hotel Recommendations → Nearby Attractions → Budget Dashboard → Final Travel Plan). Concepts covered during the internship - Machine Learning fundamentals, Neural Networks, CNNs, Transformers, Large Language Models, Hugging Face, Generative AI, and GPU computing were directly applied while building and refining these modules.

Results

The completed system successfully generates personalized travel recommendations, builds travel itineraries, suggests hotels, identifies nearby attractions, and estimates travel budgets. Using transformer-based semantic search rather than plain keyword matching allowed the system to understand the context and intent behind user preferences, which improved the relevance of the recommendations produced. The final platform delivers a complete tourism planning experience through a single, user-friendly Streamlit interface, removing the need to consult multiple separate Website.



Explore Your Journey

Review recommendations, customize your lodging, inspect maps, and review financial estimations

QUERY INTERESTS "beaches" STYLE Solo TRAVELERS 1 Person(s) DAYS 3 Days TARGET DESTINATION Calangute

Overview & Inspector

Itinerary Planner

Nearby Attractions

Budget Summary

Recommended Destinations

Our neural network mapped these locations based on your interest keywords. Hover for an immersive preview.

Whispering Calangute... 71.0% Match

Calangute, Goa
Nature 4.6

Located in the heart of Calangute, this beautiful nature destination is a key highlight of Goa tourism. This notable...

Cloudy Calangute Wil... 70.5% Match

Calangute, Goa
Nature 4.7

Situated in a scenic setting in Calangute, this nature attraction offers travelers an unforgettable local experience. This...

Scenic Beach of Muss... 69.9% Match

Mussoorie, Uttarakhand
Nature 4.8

Situated in a scenic setting in Mussoorie, this nature attraction offers travelers an unforgettable local experience. This...

Discover Lodging in Calangute

Max Price Per Night (INR)

8000

Minimum Hotel Rating

3.5

Search Amenities or Names

e.g., WiFi, Pool, Spa

Showing top 9 matching hotels:

Taj Chalet Calangute 53

₹7,482 / nt

Calangute | 5.0 KM
5.0 1040 reviews

Premium luxury accommodation in Calangute. Features world-class dining, customized guest services, elegant architecture, and a highly scenic...

Choose

Taj Villas Calangute 25

₹7,336 / nt

Calangute | 7.1 KM
5.0 179 reviews

Premium luxury accommodation in Calangute. Features world-class dining, customized guest services, elegant architecture, and a highly scenic...

Choose

Saffron Stays Stays Calang...

₹5,609 / nt

Calangute | 8.5 KM
4.9 1089 reviews

Premium luxury accommodation in Calangute. Features world-class dining, customized guest services, elegant architecture, and a highly scenic...

Choose

Ritz Castle Calangute 44

₹6,799 / nt

Calangute | 7.2 KM
4.9 1803 reviews

Premium luxury accommodation in Calangute. Features world-class dining, customized guest services, elegant architecture, and a highly scenic...

Paradise Resort Calangute 32

₹5,393 / nt

Calangute | 1.4 KM
4.9 116 reviews

Premium luxury accommodation in Calangute. Features world-class dining, customized guest services, elegant architecture, and a highly scenic...

Choose

Radisson Suites Calangute 54

₹7,072 / nt

Calangute | 8.1 KM
4.8 1393 reviews

Premium luxury accommodation in Calangute. Features world-class dining, customized guest services, elegant architecture, and a highly scenic...

Choose

Conclusion

SmartTrip AI demonstrates the practical application of Artificial Intelligence, Natural Language Processing, semantic search, and recommendation techniques to simplify tourism planning. By integrating destination recommendations, itinerary generation, hotel suggestions, nearby attraction discovery, and budget estimation into a single platform, the project transforms a fragmented and time-consuming planning process into a more convenient and guided experience.

The use of a pre-trained Transformer model and semantic similarity enables the system to provide more relevant recommendations based on user preferences and natural-language queries. Overall, the project successfully applies the concepts learned during the AI GPU Summer Internship to a real-world tourism problem and highlights the potential of AI-driven solutions to make travel planning more personalized, efficient, and user-friendly.

Reference

1. Hugging Face, sentence-transformers/all-MiniLM-L6-v2 Model Card, Available at: <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>
2. Pandas Development Team, pandas Documentation, Available at: <https://pandas.pydata.org/docs/>
3. NumPy Developers, NumPy Documentation, Available at: <https://numpy.org/doc/stable/>
4. Kaggle, Datasets and Resources for Machine Learning, Available at: <https://www.kaggle.com/>
5. Streamlit, "Streamlit Documentation", Python framework for developing data and machine learning applications.